Supplementary Information (Online Resource 7)

Article title: How Teams Decide Human Oversight in AI-Powered Systems: A Mixed-Methods Study

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Qualitative evaluation results

A. Qualitative evaluation

The qualitative research included interviews with five experts in human factors engineering. Table 1 outlines the demographic profiles of the experts interviewed.

Participant #	Expertise	Years of experience	Domain and context
1	Expert and consultant in human-machine interaction	>20	Military
2	Human-machine interaction designer in systems that include AI components	>10	Safety critical
3	Expert in Human Factors Engineering with a strong background in systems engineering, human-machine interaction, and autonomous systems.	>20	Autonomous systems, robotics
4	AI system development expert	>10	Public-sector safety, surveillance
5	Data science expert	>20	Military, autonomous systems, swarms

Table 1 – Participants demography

During these interviews, we presented the theory and invited the experts to evaluate it and suggest possible improvements. To facilitate richer feedback, we also prepared a set of complementary questions designed to prompt further reflection and critique, including:

1 – How effectively does the initial phase of this framework support capturing real-world operational context and user diversity? What additional contextual elements (e.g., cognitive workload, environmental stressors) should be included?

2 – Does the framework capture all relevant oversight dimensions (e.g., control, transparency, trust, intervention readiness)?

3 – What user-centered design principles should guide the implementation of oversight mechanisms?

4 – What evaluation methods (e.g., simulation, cognitive walkthroughs, usability testing) are most effective for refining oversight mechanisms?

5 – Have you used structured methods (e.g., MCDA, WWW models, workshops) to resolve trade-offs in past projects?

We conducted the deductive thematic analysis [Proudfoot], using Atlas.ti, on the interviews transcripts to identify key themes corresponding to each stage of the framework. The following sections present the main themes extracted from the interviews, organized by framework stages. We marked each theme extracted from the interviews as fully or partially covered in the original framework or as a new theme that needs to be added to the framework.

Legends:

- **Theme** – A key insight or recurring idea derived from expert interviews.
- **Coverage** – Indicates whether and how the theme is currently addressed in the existing framework.
- **New** – Highlights a novel decision point or activity suggested by the theme that should be incorporated into the framework.

1) Define system context & constraints

Theme	Coverage	New
User Role Differentiation	Partial	+
Domain and Risk Assessment	Full	
Ethical and Regulatory Constraints	Full	
Trust and Perception	Partial	+
Environmental constraints	Partial	+
Domain specificity	Full	
Risk factors	Full	
Perceived responsibility and accountability	Partial	+
User motivation	None	+

The following provides insight into new themes uncovered during the interviews:

Differentiation of User Roles: It is essential to distinguish between types of users, such as service recipients, system operators, and proactive users, as each role entails different oversight requirements. A one-size-fits-all approach to oversight is insufficient; involvement must be tailored to the user's role and context. Quote – "*Each user type may require a different level of involvement and oversight. For example, a system operator or a professional (like a doctor) may need a higher degree of understanding and control compared to a service recipient*."

Trust and Perception: Users' willingness to engage with the system is strongly shaped by psychological factors, including trust, prior experience, and perceived system reliability. These elements directly impact on how much oversight users are comfortable exercising. Quote – "*It doesn't matter how reliable the system actually is, what matters is how reliable the user perceives it to be.*"

Environmental Constraints: The operational setting, such as high-stress or high-tempo environments like battlefields or traffic control centers, can limit the feasibility of active oversight and situational awareness, necessitating adaptive system support. Quote – "*The cognitive burden placed on users (e.g., through information overload, complex interfaces, or frequent decision points) is a major consideration. Systems should aim to reduce unnecessary cognitive load while ensuring users have sufficient situational awareness to supervise and intervene if needed*."

Responsibility and Accountability: Effective oversight relies not only on available tools, but also on whether users have or feel a sense of responsibility for system outcomes. When accountability is unclear or diminished, users may become passive and overly reliant on automation, even when intervention is possible. Quote – "*Ethical constraints and regulatory requirements (such as the need for accountability and auditability) often dictate minimum levels of human oversight.*"

User Motivation: High levels of intrinsic or extrinsic motivation (e.g., ethical responsibility, mission urgency) can drive users to maintain engagement with the system despite usability challenges. Motivation thus acts as a critical enabler of sustained oversight behavior. Quote – "*Military operators succeed even when the interface is poor, motivation is high."*

2) *Establish oversight spectrum*

Theme	Coverage	New
Clarifying Levels of Automation (LOA):	Partial	+

Clarifying Levels of Automation (LOA): The traditional HITL/HOTL/HOOTL taxonomy provides an overly simplistic view of human roles in complex systems. There is a growing need for more precise definitions and transparent communication of automation levels, enabling users to clearly understand their expected involvement and responsibilities within the system. Quote – "*If in the past people talked about 10 levels, like Sheridan and Verplank [Sheridan], today we understand there could be 100 possible options in between. Once you talk about which routines are automated, the space of possibilities is much broader.*"

3) *Implement dynamic oversight mechanism*

Theme	Coverage	New
Situational awareness as a prerequisite	Partial	+
Explainability aligned with cognitive load	Partial	+
Monitoring and Intervention Tools	Full	
Bidirectional Interaction – feedback by the users	Full	
Captive vs. voluntary users	None	+
Mitigating over-trust \ Automation Fatigue	Partial	+
Adaptive interface design	Partial	+
Realtime vs. Retrospective decision oversight	None	+
Controlled switching between LOA	None	+
Implicit explainability	None	+

The thematic analysis of expert interviews revealed a strong emphasis on the need for clear, specific, and actionable definitions of human oversight mechanisms, rather than broad or vague categorizations. This stage of the framework was significantly enriched by novel insights, highlighting the diverse ways in which oversight can and should be operationalized in AI-powered systems. The following themes emerged:

Situational awareness as a prerequisite: Effective oversight begins with the user maintaining an understanding of the system's current actions and status, even under high levels of automation. Quote – "*You cannot prevent the user from having situational awareness. The system must support it, even if most processes are automated.*".

Explainability aligned with cognitive load: Explanations should be context-sensitive and aligned with the user's expertise and cognitive capacity, avoiding both oversimplification and unnecessary complexity. Quote – "*Users must maintain situational awareness relevant to their tasks. Systems should be designed to support the user's mental model, helping them keep track of what the system is doing and why, without overwhelming them with details*".

Captive vs. voluntary users: In environments like military operations, where users are obligated to use the system, oversight can be mandated. In contrast, civilian systems must earn user trust to achieve active oversight. Quote – *"Ordinary users either abuse the system (like with Tesla autopilot) or abandon it*.".

Mitigating over-trust: Excessive trust in automated systems can lead to disengagement and automation complacency, which is particularly risky in safety-critical settings. Oversight mechanisms should counterbalance this tendency. Quote – "*Evaluation of good oversight should focus on avoidance of blind trust or disengagement*".

Adaptive interface design: Interfaces should be configurable, supporting user-specific alerting, control preferences, and layered transparency, offering simple explanations upfront, with optional access to deeper system logic. Quote – "*Systems should be designed to support the user's mental model, helping them keep track of what the system is doing and why, without overwhelming them with details."*

Real-time vs. Retrospective decision oversight: It is crucial to distinguish between decisions that require real-time explanation (e.g., during a mission) and those that can be reviewed post hoc, allowing a balance between operational efficiency and accountability. Quote – "*In many cases, oversight is not only real-time, investigation and feedback can occur after*."

Controlled switching between LOA: The ability to switch between automation levels must be governed by clear, role-based permissions to prevent misuse while supporting flexibility and contextual needs. Quote – "*The question of who can switch control levels is very important, just the user? someone else? a third party?*".

Implicit explainability: In many situations, users do not rely on explicit, detailed model-level explanations. Rather, their ability to form shared mental models depends on the system behaving consistently and transparently. When automation becomes unpredictable or opaque, users struggle to understand its actions, weakening their trust and oversight. Quote – "*We often talk about building shared mental models... Most of the time, you insert and remove the human in emergency situations when automation collapses. But if the user doesn't understand what automation is doing, because it's inconsistent or opaque, then they lose situational awareness*."

4) Reveal contradictions

Theme	Coverage	New
Oversight vs. Performance	Full	
Transparency vs. Complexity	Partial	+
Trust vs. Automation	Partial	+
Performance vs. explainability	Full	
Usability vs. oversight depth	Full	

Transparency vs. Complexity: When users repeatedly rely on automated outputs without meaningful involvement, they may develop automation complacency and become less likely to detect errors or anomalies. Over time, this may also lead to skill decay, reducing users' readiness to intervene effectively when human judgment is required. Quote – "*For example, automation may reduce cognitive load but also erode user vigilance.* ", "*Essential to prevent skill loss, confusion, or complacency.*"

Trust vs. Automation: When systems are overly automated, users may become passive, leading to reduced situational awareness, weakened oversight and even reduced skills in system operation. Quote – "*user trust is a byproduct of good oversight and UX.*", "*If the user hasn't intervened in a while, they may suffer from skill decay or complacency.*"

5) Negotiate trade-offs

Theme	Coverage	New
Use of Structured Tools	Full	

a) Evaluate and iterate

Theme	Coverage	New
Usability and Trust as Evaluation Metrics	Full	
User-centered negotiation	Partial	+
Objective and subjective UX metrics	Partial	+
Iterative usability testing	Partial	+
Post-deployment evaluation	Full	
Need for continual adaptation	Full	
Graduated trust	Partial	+

The thematic analysis of the evaluation and iteration phase revealed a strong need to go beyond general methodological references. Participants emphasized the importance of defining clear, specific, and actionable evaluation tasks that align with system goals and user needs. As a result, this phase of the framework was significantly expanded based on the following insights:

User-Centered Negotiation: Evaluation should include usability tests and structured feedback to assess how much control or explanation users can realistically manage without compromising system performance. This fosters a negotiation between usability and functional complexity. Quote – "*We define objective metrics like task completion, error rate, and task time... and we also use subjective tools like the SUS (System Usability Scale) questionnaire to measure usability*.".

Objective and Subjective UX Metrics: Evaluation must incorporate both quantitative (e.g., task success rate, error counts) and qualitative (e.g., perceived ease of use, system usability scores) measures to holistically assess user experience and oversight effectiveness.

Iterative Usability Testing: Continuous experimentation, such as A/B testing, scenario-based trials, or interactive prototyping, is essential to determine which oversight mechanisms and interface designs yield the best outcomes. Quote – "*iterative user testing is necessary to ensure that users can effectively interact with and oversee the system as designed*."

Graduated Trust: Trust in AI systems develop progressively through repeated exposure to consistent, transparent behavior. Early phases should include stronger oversight, with the option to reduce it as the system proves reliable and users gain confidence. Quote – "*Maybe at first you give the user more updates, then later let them choose whether to keep seeing them or not."*